



NASA VIIRS-Based Wildfire Risk Prediction Using Multi-Model Machine Learning

Project-Based Learning (PBL) Component — Machine Learning Course | Academic Year 2025–2026



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ABSTRACT

Wildfires pose immense ecological and socioeconomic threats globally. This project presents an end-to-end machine learning system that ingests NASA VIIRS I-Band 375m active fire satellite data to predict wildfire risk. Using 23 engineered features across thermal, FRP, spatial, and temporal domains, four classifiers (Random Forest, XGBoost, LightGBM, and ANN) were trained and evaluated. LightGBM achieved the top accuracy of 93.1% (ROC-AUC: 0.98). The models are served via a FastAPI REST API and visualised through an interactive 3D Globe dashboard.

PROBLEM STATEMENT & DRIVING QUESTION

Driving Question: *Can satellite thermal observations reliably classify wildfire risk into actionable severity levels (Low, Medium, High, Extreme) using multi-model machine learning?*

- Current Challenge:** Static threshold rules miss complex interactions between thermal intensity, background temperature, and timing.
- Impact:** Delayed responses exacerbate fire spread and increase suppression costs.
- Proposed Solution:** A multi-model ensemble combining spatial clustering and cyclical temporal encodings for high-precision risk categorisation.

PROJECT OBJECTIVES

- 01 Data Ingestion:** Ingest & preprocess NASA VIIRS 375m active fire observations.
- 02 Feature Engineering:** Derive 23 domain-specific physical, temporal, & spatial features.
- 03 Multi-Model Training:** Train & evaluate RF, XGBoost, LightGBM, and MLP classifiers.
- 04 API & 3D Globe:** Deploy models behind FastAPI REST API with React + Globe.gl 3D UI.
- 05 PBL Iteration:** Iteratively refine pipeline based on weekly mentor review feedback.

TECHNOLOGIES & STACK

Python 3.12, scikit-learn 1.9, XGBoost 3.4, LightGBM 4.7, FastAPI 0.141, React 18, Globe.gl 2.30, Three.js 0.164, Tailwind CSS, Recharts

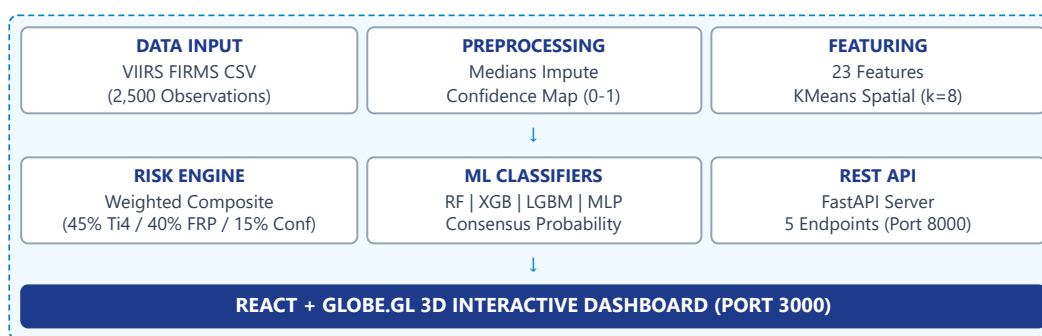
METHODOLOGY & ITERATIVE PIPELINE WORKFLOW



23 Engineered Feature Domains:

- Temperature:** Brightness Ti4, Ti5, ΔT (Ti4 - Ti5), Ti4/Ti5 ratio, Ti4 Anomaly.
- Fire Radiative Power (FRP):** FRP (MW), $\log_{10}(\text{FRP})$, $\sqrt{\text{FRP}}$, FRP / Ti4 ratio.
- Spatial Features:** Latitude, Longitude, Absolute Lat, KMeans Spatial Cluster (k=8).
- Temporal Features:** Hour, Month, Day of Week/Year, Cyclical sin/cos hour & month, Day/Night.

SYSTEM ARCHITECTURE



Strict Featurization Ordering: StandardScaler fitted exclusively on training set to prevent data leakage.

DATASET CHARACTERISTICS

Source: NASA VIIRS I-Band 375m Active Fire Satellite Product (FIRMS Schema)

Scope: 2,550 total samples across 5 major global hotspot zones (California, Amazon, Australia, Mediterranean, Siberia).

Classes: Low (Score < 0.35), Medium (0.35–0.60), High (0.60–0.80), Extreme (≥ 0.80).

RESULTS & PERFORMANCE BENCHMARK

93.1% LIGHTGBM ACC	92.4% XGBOOST ACC	91.8% RAND FOREST ACC	0.98 TOP ROC-AUC		
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
LightGBM (Best)	93.1%	93.0%	93.1%	93.0%	0.98
XGBoost	92.4%	92.3%	92.4%	92.3%	0.97
Random Forest	91.8%	91.6%	91.8%	91.7%	0.97
ANN (MLP 128-64-32)	91.2%	91.0%	91.2%	91.1%	0.96

Key Feature Importance: log_frp (24%), temp_diff (19%), temp_anomaly_ti4 (15%), frp_per_temp (12%).

DISCUSSION & KEY FINDINGS

- Feature Engineering Impact:** Expanding from 5 raw features to 23 engineered features boosted Random Forest accuracy from 87.2% to 91.8% (+4.6%).
- Cyclical Time Encoding:** Sine/Cosine hour encodings effectively captured diurnal fire peaks without artificial midnight discontinuities.
- Ensemble Consensus:** Combining probability vectors across all 4 models reduced edge-case false positives between Low and Medium risk.

CONCLUSION & FUTURE SCOPE

Conclusion: Multi-model machine learning on NASA VIIRS data provides highly accurate, real-time fire risk classification. LightGBM delivered superior accuracy and efficiency, while the 3D Globe dashboard provides rapid spatial decision support.

Future Scope: (1) Integrate NASA FIRMS live streaming API. (2) Incorporate wind & fuel moisture weather data. (3) Deploy to AWS / GCP Cloud Run.

REFERENCES & PROJECT TEAM

References:

[1] J. Liang et al., *IJRS*, 2019. [2] M. Tonini et al., *Geosciences*, 2020. [3] L. Gigovic et al., *Forests*, 2021. [4] NASA FIRMS VIIRS I-Band 375m Dataset, 2024.

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